

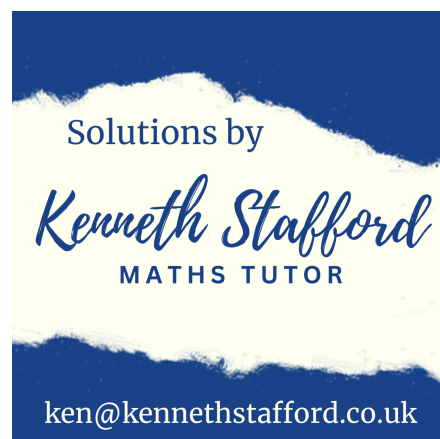
Name: _____

GCSE (1 - 9)

Standard Form
(Higher Paper)



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These questions are compiled from *released* [Pearson Edexcel GCSE maths exam papers](#)

They contain exam questions from up until the **Summer 2023** exam series

More recent exam papers do exist but are withheld for mock exams

As new exam papers are released these worksheets will gradually be updated

The **most current** version of these worksheets will always be available at: [kennethstafford.co.uk](#)

These worksheets were made using [examq.co.uk](#)

Question	Where the question appeared
1	Edexcel, GCSE, Autumn 2017, Paper 3H, Q7
2	Edexcel, GCSE, Autumn 2018, Paper 2H, Q7
3	Edexcel, GCSE, Autumn 2019, Paper 2H, Q7
4	Edexcel, GCSE, Autumn 2019, Paper 3H, Q7
5	Edexcel, GCSE, Autumn 2020, Paper 1H, Q9
6	Edexcel, GCSE, Autumn 2020, Paper 3H, Q10
7	Edexcel, GCSE, Summer 2017, Paper 1H, Q8
8	Edexcel, GCSE, Summer 2017, Paper 2H, Q10
9	Edexcel, GCSE, Summer 2018, Paper 3H, Q9
10	Edexcel, GCSE, Summer 2019, Paper 2H, Q8
11	Edexcel, GCSE, Summer 2019, Paper 3H, Q11

More questions and solutions: [www.kennethstafford.co.uk](#)

Standard Form (Higher Paper) (Questions from Edexcel GCSE Past Papers)

1

Work out $(13.8 \times 10^7) \times (5.4 \times 10^{-12})$
Give your answer as an ordinary number.

0.0007452

(Total for Question 7 is 2 marks)

[Edexcel, GCSE, Autumn 2017, Paper 3H, Q7]

2

(a) Write the number 0.00008623 in standard form.

8.623×10^{-5}

(1)

(b) Work out $\frac{3.2 \times 10^3 + 5.1 \times 10^{-2}}{4.3 \times 10^{-4}}$

Give your answer in standard form, correct to 3 significant figures.

7,441 979.07

7.44×10^6

(2)

(Total for Question 7 is 3 marks)

[Edexcel, GCSE, Autumn 2018, Paper 2H, Q7]

Standard Form (Higher Paper) (Questions from Edexcel GCSE Past Papers)

3

Work out $(3.42 \times 10^{-7}) \div (7.5 \times 10^{-6})$

Give your answer in standard form.

$$= 0.0426$$

$$4.56 \times 10^{-2}$$

(Total for Question 7 is 2 marks)

[Edexcel, GCSE, Autumn 2019, Paper 2H, Q7]

Standard Form (Higher Paper) (Questions from Edexcel GCSE Past Papers)

4

(a) Write 32 460 000 in standard form.

$$3.246 \times 10^7$$

(1)

(b) Write 4.96×10^{-3} as an ordinary number.

$$0.00496$$

(1)

Asma was asked to compare the following two numbers.

$$A = 6.212 \times 10^8 \quad \text{and} \quad B = 4.73 \times 10^9$$

She says,

“6.212 is bigger than 4.73 so A is bigger than B .”

(c) Is Asma correct?

You must give a reason for your answer.

No.

$$10^8 < 10^9$$

$$62,120,000 < 4,730,000,000$$

(1)

(Total for Question 7 is 3 marks)

[Edexcel, GCSE, Autumn 2019, Paper 3H, Q7]

Standard Form (Higher Paper) (Questions from Edexcel GCSE Past Papers)

5

Write these numbers in order of size.
Start with the smallest number.

$$6.72 \times 10^5$$

$$672,000$$

$$67.2 \times 10^{-4}$$

$$0.00672$$

$$672 \times 10^4$$

$$6720000$$

$$0.000672$$

$$0.000672$$

$$0.000672$$

$$67.2 \times 10^{-4}$$

$$6.72 \times 10^5$$

$$672 \times 10^4$$

(Total for Question 9 is 2 marks)

[Edexcel, GCSE, Autumn 2020, Paper 1H, Q9]

Standard Form (Higher Paper) (Questions from Edexcel GCSE Past Papers)

6

A person's heart beats approximately 10^5 times each day.

A person lives for approximately 81 years.

- (a) Work out an estimate for the number of times a person's heart beats in their lifetime.
Give your answer in standard form correct to 2 significant figures.

$$10^5 \times 365 \times 81$$

$$10^5 \times 400 \times 80 = 32000 \times 10^5 \\ = 3.2 \times 10^9$$

$$\frac{3.2 \times 10^9}{(2)}$$

2×10^{12} red blood cells have a total mass of 90 grams.

- (b) Work out the average mass of 1 red blood cell.
Give your answer in standard form.

$$\frac{90}{2 \times 10^{12}} = 4.5 \times 10^{-11}$$

$$\frac{4.5 \times 10^{-11}}{(2)} \text{ grams}$$

(Total for Question 10 is 4 marks)

[Edexcel, GCSE, Autumn 2020, Paper 3H, Q10]

Standard Form (Higher Paper) (Questions from Edexcel GCSE Past Papers)

7

- (a) Write 7.97×10^{-6} as an ordinary number.

0.00000797

(1)

- (b) Work out the value of $(2.52 \times 10^5) \div (4 \times 10^{-3})$
Give your answer in standard form.

$$\begin{array}{r} 0.63 \\ 4 \overline{) 2.52} \end{array}$$

$$0.63 \times 10^{5-(-3)}$$

$$0.63 \times 10^8$$

$$6.3 \times 10^7$$

(2)

(Total for Question 8 is 3 marks)

[Edexcel, GCSE, Summer 2017, Paper 1H, Q8]

8

The table shows some information about eight planets.

Planet	Distance from Earth (km)	Mass (kg)
Earth	0	5.97×10^{24}
Jupiter	6.29×10^8	1.898×10^{27}
Mars	7.83×10^7	6.42×10^{23}
Mercury	9.17×10^7	3.302×10^{23}
Neptune	4.35×10^9	1.024×10^{26}
Saturn	1.28×10^9	5.68×10^{26}
Uranus	2.72×10^9	8.683×10^{25}
Venus	4.14×10^7	4.869×10^{24}

(a) Write down the name of the planet with the greatest mass.

Jupiter

(1)

(b) Find the difference between the mass of Venus and the mass of Mercury.

$$4.869 \times 10^{24} - 3.302 \times 10^{23} =$$

$$4.5388 \times 10^{24} \text{ kg}$$

(1)

Nishat says that Neptune is over a hundred times further away from Earth than Venus is.

(c) Is Nishat right?

You must show how you get your answer.

$$4.35 \times 10^9 \div 4.14 \times 10^7 = 105.07$$

Yes more than 100

$$105.07 > 100$$

(2)

(Total for Question 10 is 4 marks)

[Edexcel, GCSE, Summer 2017, Paper 2H, Q10]

9

$$T = \sqrt{\frac{w}{d^3}}$$

$$w = 5.6 \times 10^{-5}$$

$$d = 1.4 \times 10^{-4}$$

(a) Work out the value of T .

Give your answer in standard form correct to 3 significant figures.

$$\sqrt{\frac{5.6 \times 10^{-5}}{(1.4 \times 10^{-4})^3}} = 6324.55$$

$$T = \frac{6.32 \times 10^3}{(2)}$$

 w is increased by 10% d is increased by 5%

Lottie says,

“The value of T will increase because both w and d are increased.”

(b) Lottie is wrong.

Explain why.

$$\sqrt{\frac{1.1}{(1.05)^3}} = 0.974$$

$$1 - 0.974 = 0.026$$

Overall 2.6% reduction.

(2)

(Total for Question 9 is 4 marks)

[Edexcel, GCSE, Summer 2018, Paper 3H, Q9]

Standard Form (Higher Paper) (Questions from Edexcel GCSE Past Papers)

10

(a) Write 0.00562 in standard form.

$$\frac{5.62 \times 10^{-3}}{(1)}$$

(b) Write 1.452×10^3 as an ordinary number.

$$\frac{1452}{(1)}$$

(Total for Question 8 is 2 marks)

[Edexcel, GCSE, Summer 2019, Paper 2H, Q8]

11

In May 2019, the distance between Earth and Mars was 3.9×10^7 km.

In May 2019, a signal was sent from Earth to Mars.

Assuming that the signal sent from Earth to Mars travelled at a speed of 3×10^5 km per second,

(a) how long did the signal take to get to Mars?

$$\text{time} = \frac{\text{distance}}{\text{speed}} = \frac{3.9 \times 10^7}{3 \times 10^5} = 130$$

$$\frac{130}{(2)} \text{ seconds}$$

The speed of the signal sent from Earth to Mars in May 2019 was actually less than 3×10^5 km per second.

(b) How will this affect your answer to part (a)?

It will take longer.

(1)

(Total for Question 11 is 3 marks)

[Edexcel, GCSE, Summer 2019, Paper 3H, Q11]